

WR-I Mathematical Spine v0.1

Response-Defined World Spaces:
Finite-Resolution Equivalence and Identifiability of Global Realizations

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Research Programme & Community

6 October 2026

Abstract

This document fixes the mathematical spine for the first technical paper of the World Realizability & Epistemic Horizons (WREH) programme. The basic object is a class of admissible global realizations equipped with a family of admissible observation protocols and their response maps. Exact response equality induces a quotient world space. We prove the canonical quotient representation, factorization of identifiable quantities, monotonicity under protocol refinement, factorization through gauge orbits, and a criterion for when response non-uniqueness is exhausted by gauge redundancy. We also separate exact equivalence from finite-tolerance indistinguishability, for which transitivity generally fails. Smooth and infinite-dimensional examples show how refinement can remove genuine non-identifiability or leave persistent operational ambiguity. The framework is deliberately ontologically neutral: response-equivalent realizations are not declared physically identical, and no claim is made that observation creates, selects, or modifies a world.

1 Scope and claim ceiling

WR-I is an abstract response-identifiability paper. The term “world” means an element of a declared class of global realizations. It does not, by itself, mean a cosmological universe or a physically realized ontology.

The paper does *not* claim that:

- response-equivalent realizations are ontologically identical;
- underlying reality does not exist;
- measurement creates or selects a unique world;
- finite-dimensional response walls are spacetime boundaries;
- protocol refinement is physical time, dynamics, or renormalization flow.

2 World-response structures

Definition 2.1 (World-response structure). *A world-response structure is a triple*

$$\mathbf{W} = (\mathcal{W}, \mathcal{P}, \{R_P\}_{P \in \mathcal{P}}),$$

where $\mathcal{W}$ is a set of declared admissible global realizations, $\mathcal{P}$ is a set of admissible observation protocols, and each protocol P has a response map

$$R_P : \mathcal{W} \rightarrow Y_P$$

into a protocol-dependent response space Y_P .

Definition 2.2 (Protocol bundle and joint response). For $\mathcal{A} \subseteq \mathcal{P}$, define

$$Y_{\mathcal{A}} := \prod_{P \in \mathcal{A}} Y_P$$

and the joint response

$$R_{\mathcal{A}} : \mathcal{W} \rightarrow Y_{\mathcal{A}}, \quad R_{\mathcal{A}}(W) := (R_P(W))_{P \in \mathcal{A}}.$$

For the empty bundle, $Y_{\emptyset}$ is the one-point set.

Definition 2.3 (Exact response equivalence). For a protocol bundle $\mathcal{A}$, define

$$W_1 \sim_{\mathcal{A}} W_2 \iff R_{\mathcal{A}}(W_1) = R_{\mathcal{A}}(W_2).$$

Equivalently,

$$R_P(W_1) = R_P(W_2) \text{ for every } P \in \mathcal{A}.$$

Definition 2.4 (Response-defined world space). The response-defined world space for $\mathcal{A}$ is the quotient set

$$\mathcal{Q}_{\mathcal{A}} := \mathcal{W} / \sim_{\mathcal{A}}.$$

Its canonical quotient map is denoted

$$q_{\mathcal{A}} : \mathcal{W} \rightarrow \mathcal{Q}_{\mathcal{A}}, \quad q_{\mathcal{A}}(W) = [W]_{\mathcal{A}}.$$

3 Canonical quotient representation

Proposition 3.1 (Response quotient representation). The relation $\sim_{\mathcal{A}}$ is an equivalence relation. Moreover, there is a unique bijection

$$\overline{R}_{\mathcal{A}} : \mathcal{Q}_{\mathcal{A}} \longrightarrow \text{Im } R_{\mathcal{A}}$$

such that

$$R_{\mathcal{A}} = \overline{R}_{\mathcal{A}} \circ q_{\mathcal{A}}.$$

Proof. Equality in $Y_{\mathcal{A}}$ is reflexive, symmetric, and transitive. Since $\sim_{\mathcal{A}}$ is the pullback of equality along $R_{\mathcal{A}}$, it is an equivalence relation.

Define

$$\overline{R}_{\mathcal{A}}([W]_{\mathcal{A}}) := R_{\mathcal{A}}(W).$$

If $[W]_{\mathcal{A}} = [W']_{\mathcal{A}}$, then $W \sim_{\mathcal{A}} W'$, hence $R_{\mathcal{A}}(W) = R_{\mathcal{A}}(W')$. Therefore the definition is well posed. The displayed factorization follows immediately.

The map is surjective onto $\text{Im } R_{\mathcal{A}}$ by definition of the image. If

$$\overline{R}_{\mathcal{A}}([W]_{\mathcal{A}}) = \overline{R}_{\mathcal{A}}([W']_{\mathcal{A}}),$$

then $R_{\mathcal{A}}(W) = R_{\mathcal{A}}(W')$, so $W \sim_{\mathcal{A}} W'$ and $[W]_{\mathcal{A}} = [W']_{\mathcal{A}}$. Hence $\overline{R}_{\mathcal{A}}$ is injective. Uniqueness follows because $q_{\mathcal{A}}$ is surjective. $\square$

Remark 3.2. As a bare set, $\mathcal{Q}_{\mathcal{A}}$ contains exactly the information present in the joint response image. Additional geometry, topology, measure, smooth structure, or stratification on $\mathcal{Q}_{\mathcal{A}}$ requires additional assumptions and is not automatic.

4 Identifiable quantities

Definition 4.1 (Identifiable quantity). *Let $F : \mathcal{W} \rightarrow Z$. We say that F is $\mathcal{A}$ -identifiable if*

$$W_1 \sim_{\mathcal{A}} W_2 \implies F(W_1) = F(W_2).$$

Thus F is constant on each response fibre.

Proposition 4.2 (Identifiability factorization). *A map $F : \mathcal{W} \rightarrow Z$ is $\mathcal{A}$ -identifiable if and only if there exists a unique map*

$$\bar{F} : \mathcal{Q}_{\mathcal{A}} \rightarrow Z$$

such that

$$F = \bar{F} \circ q_{\mathcal{A}}.$$

Proof. Assume first that F is $\mathcal{A}$ -identifiable. Define

$$\bar{F}([W]_{\mathcal{A}}) := F(W).$$

If $[W]_{\mathcal{A}} = [W']_{\mathcal{A}}$, identifiability gives $F(W) = F(W')$, so $\bar{F}$ is well defined. The factorization follows directly.

Conversely, if $F = \bar{F} \circ q_{\mathcal{A}}$ and $W_1 \sim_{\mathcal{A}} W_2$, then $q_{\mathcal{A}}(W_1) = q_{\mathcal{A}}(W_2)$, hence $F(W_1) = F(W_2)$. Uniqueness again follows from surjectivity of $q_{\mathcal{A}}$. $\square$

Remark 4.3. *This proposition gives the operational content of identifiability: every quantity determined by the declared protocol family must descend to the response quotient.*

5 Protocol refinement

Definition 5.1 (Refinement order). *For protocol bundles $\mathcal{A}, \mathcal{B} \subseteq \mathcal{P}$, write*

$$\mathcal{A} \preceq \mathcal{B} \iff \mathcal{A} \subseteq \mathcal{B}.$$

Proposition 5.2 (Refinement monotonicity). *If $\mathcal{A} \subseteq \mathcal{B}$, then*

$$W_1 \sim_{\mathcal{B}} W_2 \implies W_1 \sim_{\mathcal{A}} W_2,$$

and therefore

$$[W]_{\mathcal{B}} \subseteq [W]_{\mathcal{A}} \quad \text{for every } W \in \mathcal{W}.$$

There is a unique surjection

$$\pi_{\mathcal{B}\mathcal{A}} : \mathcal{Q}_{\mathcal{B}} \rightarrow \mathcal{Q}_{\mathcal{A}}, \quad [W]_{\mathcal{B}} \mapsto [W]_{\mathcal{A}},$$

satisfying

$$q_{\mathcal{A}} = \pi_{\mathcal{B}\mathcal{A}} \circ q_{\mathcal{B}}.$$

For $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{C}$,

$$\pi_{\mathcal{C}\mathcal{A}} = \pi_{\mathcal{B}\mathcal{A}} \circ \pi_{\mathcal{C}\mathcal{B}}.$$

Proof. Equality of all responses indexed by $\mathcal{B}$ implies equality of the subfamily indexed by $\mathcal{A}$. Hence $\sim_{\mathcal{B}}$ refines $\sim_{\mathcal{A}}$.

If $[W]_{\mathcal{B}} = [W']_{\mathcal{B}}$, then $W \sim_{\mathcal{B}} W'$, and therefore $W \sim_{\mathcal{A}} W'$. Thus $\pi_{\mathcal{B}\mathcal{A}}$ is well defined. It is surjective because every $\mathcal{A}$ -class has a representative W , and the $\mathcal{B}$ -class of that same representative maps to it. The commuting identity and the composition law are immediate from the definition. $\square$

Remark 5.3. *The family $(\mathcal{Q}_{\mathcal{A}}, \pi_{\mathcal{B}\mathcal{A}})$ is the natural input for the global-realizability questions of WR-II. WR-I does not assume that any inverse limit has a physically meaningful interpretation.*

6 Gauge redundancy and response non-identifiability

Definition 6.1 (Gauge action). *Let a group G act on $\mathcal{W}$. The action is response-invariant for $\mathcal{A}$ if*

$$R_P(g \cdot W) = R_P(W)$$

for every $g \in G$, $W \in \mathcal{W}$, and $P \in \mathcal{A}$.

Proposition 6.2 (Gauge factorization). *If the G -action is response-invariant for $\mathcal{A}$, then*

$$G \cdot W \subseteq [W]_{\mathcal{A}}$$

for every $W \in \mathcal{W}$. Furthermore, $R_{\mathcal{A}}$ factors uniquely through the orbit quotient:

$$\mathcal{W} \xrightarrow{\pi_G} \mathcal{W}/G \xrightarrow{\tilde{R}_{\mathcal{A}}} Y_{\mathcal{A}},$$

where

$$\tilde{R}_{\mathcal{A}}(G \cdot W) = R_{\mathcal{A}}(W).$$

Proof. For every $g \in G$, response invariance gives $R_{\mathcal{A}}(g \cdot W) = R_{\mathcal{A}}(W)$, so $g \cdot W \sim_{\mathcal{A}} W$. Thus the entire orbit is contained in the response class.

Define

$$\tilde{R}_{\mathcal{A}}(G \cdot W) := R_{\mathcal{A}}(W).$$

If W' lies in the same orbit as W , response invariance gives $R_{\mathcal{A}}(W') = R_{\mathcal{A}}(W)$; hence the map is well defined. The factorization is immediate, and uniqueness follows from surjectivity of π_G . $\square$

Proposition 6.3 (Gauge-completeness criterion). *Assume the G -action is response-invariant for $\mathcal{A}$. Then the following are equivalent:*

1. $[W]_{\mathcal{A}} = G \cdot W$ for every $W \in \mathcal{W}$;
2. the induced map

$$\tilde{R}_{\mathcal{A}} : \mathcal{W}/G \rightarrow Y_{\mathcal{A}}$$

is injective.

Proof. Assume (1). Suppose

$$\tilde{R}_{\mathcal{A}}(G \cdot W_1) = \tilde{R}_{\mathcal{A}}(G \cdot W_2).$$

Then $R_{\mathcal{A}}(W_1) = R_{\mathcal{A}}(W_2)$, so $W_1 \sim_{\mathcal{A}} W_2$. By (1), $W_2 \in [W_1]_{\mathcal{A}} = G \cdot W_1$. Thus the two orbits are equal, proving injectivity.

Conversely, assume (2). Proposition 5.2 already gives

$$G \cdot W \subseteq [W]_{\mathcal{A}}.$$

Let $W' \in [W]_{\mathcal{A}}$. Then $R_{\mathcal{A}}(W') = R_{\mathcal{A}}(W)$, so

$$\tilde{R}_{\mathcal{A}}(G \cdot W') = \tilde{R}_{\mathcal{A}}(G \cdot W).$$

Injectivity implies $G \cdot W' = G \cdot W$, hence $W' \in G \cdot W$. Therefore $[W]_{\mathcal{A}} \subseteq G \cdot W$, proving equality. $\square$

Remark 6.4. *The criterion separates representational redundancy from genuine response non-identifiability. If $\tilde{R}_{\mathcal{A}}$ is non-injective, then distinct gauge orbits remain observationally indistinguishable.*

7 Finite-resolution separation on compact physical state spaces

The previous propositions are set-theoretic. To connect exact identifiability with finite experimental design, assume now that the gauge quotient

$$X := \mathcal{W}/G$$

is a compact metric space with metric d_X , that each gauge-invariant protocol induces a continuous map

$$\tilde{R}_P : X \rightarrow Y_P$$

into a metric space (Y_P, d_P) , and that the full admissible family separates points of X :

$$x \neq y \implies \exists P \in \mathcal{P} : \tilde{R}_P(x) \neq \tilde{R}_P(y).$$

Proposition 7.1 (Finite-resolution separation theorem). *Under the assumptions above, for every $\delta > 0$ there exist a finite protocol bundle*

$$\mathcal{A}_\delta = \{P_1, \dots, P_m\} \subseteq \mathcal{P}$$

and a constant $\eta_\delta > 0$ such that

$$d_X(x, y) \geq \delta$$

implies

$$\max_{1 \leq j \leq m} d_{P_j}(\tilde{R}_{P_j}(x), \tilde{R}_{P_j}(y)) \geq \eta_\delta.$$

Thus every pair of physical states separated by at least δ in the declared metric is distinguished by one of finitely many protocols, with a uniform positive response margin.

Proof. Define the compact subset

$$K_\delta = \{(x, y) \in X \times X : d_X(x, y) \geq \delta\}.$$

For each protocol P , let

$$U_P = \{(x, y) \in K_\delta : d_P(\tilde{R}_P(x), \tilde{R}_P(y)) > 0\}.$$

Continuity of $\tilde{R}_P$ implies that U_P is open in K_δ . Because the full protocol family separates points and every pair in K_δ is distinct, the family $\{U_P\}_{P \in \mathcal{P}}$ covers K_δ . Compactness yields a finite subcover

$$U_{P_1}, \dots, U_{P_m}.$$

Now define

$$h(x, y) = \max_{1 \leq j \leq m} d_{P_j}(\tilde{R}_{P_j}(x), \tilde{R}_{P_j}(y)).$$

The function h is continuous and strictly positive on K_δ , because the selected U_{P_j} cover K_δ . By compactness,

$$\eta_\delta := \min_{(x, y) \in K_\delta} h(x, y) > 0.$$

This is the required uniform separation margin. $\square$

Remark 7.2. *The proposition does not assert that one finite protocol bundle identifies all points exactly. As $\delta \downarrow 0$, the required bundle size may grow and the margin η_δ may tend to zero. The result is therefore a finite-resolution statement, not a finite exact-reconstruction theorem.*

Remark 7.3. *Compactness, the chosen metric on X , continuity of the protocol responses, and joint point separation are substantive assumptions. In a physical application each must be independently justified; none follows merely from the word “world”.*

8 Finite tolerance is not an equivalence relation in general

Assume each Y_P carries a metric d_P , and choose positive weights w_P for which the following supremum is finite on the pairs under consideration:

$$d_{\mathcal{A}}(W_1, W_2) = \sup_{P \in \mathcal{A}} w_P d_P(R_P(W_1), R_P(W_2)).$$

For $\varepsilon > 0$, define the tolerance relation

$$W_1 \approx_{\mathcal{A}, \varepsilon} W_2 \iff d_{\mathcal{A}}(W_1, W_2) \leq \varepsilon.$$

Proposition 8.1 (Failure of transitivity at finite tolerance). *The relation $\approx_{\mathcal{A}, \varepsilon}$ is generally not transitive and therefore does not, in general, define a quotient.*

Proof. Take one scalar response and three realizations with

$$R(W_0) = 0, \quad R(W_1) = \frac{3}{4}\varepsilon, \quad R(W_2) = \frac{3}{2}\varepsilon.$$

Then

$$|R(W_0) - R(W_1)| = \frac{3}{4}\varepsilon \leq \varepsilon,$$

and

$$|R(W_1) - R(W_2)| = \frac{3}{4}\varepsilon \leq \varepsilon,$$

but

$$|R(W_0) - R(W_2)| = \frac{3}{2}\varepsilon > \varepsilon.$$

Hence

$$W_0 \approx_{\varepsilon} W_1, \quad W_1 \approx_{\varepsilon} W_2, \quad W_0 \not\approx_{\varepsilon} W_2.$$

□

Remark 8.2. *WR-I therefore reserves “equivalence class” for exact response equality unless an explicit equivalence closure or another mathematically justified coarse relation is introduced. Finite-precision objects are called tolerance fibres, response neighbourhoods, or uncertainty regions.*

9 Examples

Example 9.1 (Finite binary world with gauge and hidden physical content). *Let*

$$\mathcal{W} = \{0, 1\}^4, \quad W = (a, b, g, h).$$

Let $G = \mathbb{Z}_2$ act by flipping g and leaving (a, b, h) fixed. Define

$$R_a(W) = a, \quad R_b(W) = b, \quad R_h(W) = a \oplus h.$$

With only R_a , there are two response classes of size eight. With (R_a, R_b) , there are four classes of size four. With (R_a, R_b, R_h) , there are eight classes of size two, and each class is exactly one gauge orbit. Thus the full admissible response map is injective on $\mathcal{W}/G$.

Adding an artificial protocol $R_g(W) = g$ destroys gauge invariance. It illustrates why the admissible protocol family must be part of the model rather than an arbitrary list of mathematical readouts.

Example 9.2 (Smooth world space: refinement from genuine ambiguity to pure gauge). *Let*

$$\mathcal{W} = S_x^1 \times S_g^1,$$

and let the gauge group $G = S^1$ act by translations in the g coordinate:

$$\alpha \cdot (x, g) = (x, g + \alpha).$$

Then

$$\mathcal{W}/G \cong S_x^1.$$

Consider two gauge-invariant protocols

$$R_c(x, g) = \cos x, \quad R_s(x, g) = \sin x.$$

For $\mathcal{A} = \{R_c\}$, the induced map on the gauge quotient is

$$\tilde{R}_{\mathcal{A}} : S^1 \rightarrow [-1, 1], \quad x \mapsto \cos x.$$

It is not injective. For generic x , the response fibre contains two distinct gauge orbits, represented by x and $-x$. Thus the ambiguity is not exhausted by gauge.

For $\mathcal{B} = \{R_c, R_s\}$,

$$\tilde{R}_{\mathcal{B}} : S^1 \rightarrow S^1 \subset \mathbb{R}^2, \quad x \mapsto (\cos x, \sin x)$$

is injective. Hence

$$[W]_{\mathcal{B}} = G \cdot W$$

for every W . Protocol refinement has removed genuine response non-identifiability while leaving gauge redundancy intact.

Example 9.3 (Continuous histories and dense observations). *Let*

$$\mathcal{W} = C([0, 1], \mathbb{R})$$

be the space of continuous real-valued histories. For each $t \in [0, 1]$, define the evaluation protocol

$$P_t(f) = f(t).$$

For a finite set

$$\mathcal{A} = \{t_1, \dots, t_n\},$$

the response fibre through f is

$$[f]_{\mathcal{A}} = \{g \in C([0, 1]) : g(t_i) = f(t_i) \text{ for } i = 1, \dots, n\}.$$

Equivalently,

$$[f]_{\mathcal{A}} = f + V_{\mathcal{A}}, \quad V_{\mathcal{A}} = \{u \in C([0, 1]) : u(t_i) = 0 \text{ for all } i\}.$$

The vector space $V_{\mathcal{A}}$ is infinite-dimensional, so every finite sampling bundle leaves an infinite-dimensional ambiguity.

Now let $D \subset [0, 1]$ be dense and countable. If

$$f(t) = g(t) \quad \text{for every } t \in D,$$

continuity implies $f = g$ on all of $[0, 1]$. Hence the protocol family $\{P_t : t \in D\}$ is jointly injective on $\mathcal{W}$.

The conclusion depends essentially on the admissible realization class. If $\mathcal{W}$ were the set of all functions $[0, 1] \rightarrow \mathbb{R}$ with no continuity assumption, equality on a dense subset would not imply equality everywhere. Thus identifiability is controlled jointly by the protocol family and the admissibility conditions on $\mathcal{W}$.

10 External benchmark: anisotropic Calderón gauge

A standard benchmark is the anisotropic Calderón inverse problem. The Dirichlet-to-Neumann map for a Riemannian metric is invariant under diffeomorphisms that fix the boundary. Thus boundary-fixing diffeomorphism orbits lie inside response fibres. Rigidity results seek conditions under which equality of the boundary response implies equality up to precisely this gauge. This is a concrete domain-specific analogue of the gauge-completeness criterion.

WR-I does not claim to rediscover this phenomenon. It uses it as a control case for the abstract distinction between:

representational redundancy and genuine response non-identifiability.

11 Prior-art boundary

Prior line	Established idea	WR-I boundary
Rothenberg (1971), structural identification	Observational equivalence and identifiability of structures in parametric stochastic models; identification asks whether observable laws distinguish the underlying structure.	WR-I does not claim novelty for observational equivalence. It packages admissible <i>global realizations</i> , a protocol-indexed response family, gauge separation, and a refinement system intended to feed a later global-realizability problem.
Zhang (2026), Mathematical Identifiability Geometry	Observation-equivalence classes are treated geometrically; indistinguishability strata and unique-identification points are explicit geometric objects.	WR-I must not claim that “geometry of equivalence classes” is new. Its distinctive target is a physically admissible world-response structure and the separation of gauge, protocol refinement, and downstream global completion.
Manchak (2009), global spacetime underdetermination	Classical general relativity admits observationally indistinguishable spacetimes with different global structure.	Manchak is a domain-specific predecessor for global-world underdetermination. WR-I abstracts away from the GR-specific definition of observational access and makes the admissible protocol family explicit.
Cinti–Fano (2021), critique of Manchak	Observationally indistinguishable spacetimes may fail proposed criteria of physical reasonableness.	This supports treating the admissible realization class $\mathcal{W}_{\text{admissible}}$ as part of the mathematical input rather than using all formally constructible models.

Prior line	Established idea	WR-I boundary
Abramsky– Brandenburger (2011), contextuality	Compatible local empirical data may fail to admit a global section; global realizability can be obstructed.	This is a major predecessor for WR-II, not a novelty claim of WR-I. WREH will need to distinguish global outcome assignments from global world realizations.
Anisotropic Calderón inverse problem	Boundary responses are invariant under boundary-fixing diffeomorphisms; uniqueness is naturally formulated up to gauge.	Provides a benchmark for Propositions 5.2–5.3 and for the distinction between gauge redundancy and residual non-identifiability.

12 Novelty claim permitted at v0.1

The defensible novelty claim at this stage is architectural rather than elementary-theorem novelty:

WR-I formulates a protocol-indexed response framework whose primary state space is a declared class of admissible global realizations, separates gauge redundancy from residual response non-identifiability, organizes protocol refinement by canonical quotient maps, and connects ideal joint identifiability to finite experimental separation at any fixed nonzero resolution on compact metric gauge quotients. The resulting hierarchy is then prepared for a distinct global-realizability problem.

This claim must remain subject to further literature audit.

13 Handoff to WR-II

WR-II may ask whether a refinement family of nonempty finite-response fibres admits a globally compatible realization. In the language established here, one may study systems such as

$$\mathcal{Q}_1 \leftarrow \mathcal{Q}_2 \leftarrow \mathcal{Q}_3 \leftarrow \cdots$$

and the compatibility or failure of global completion.

The present paper does not identify:

finite consistency with global existence.

Compactness assumptions, inverse/projective limits, sheaf-theoretic obstructions, and explicit counterexamples belong to the next paper.

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